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Section 6: About the Burner

A. Introduction

Thank you for purchasing a Series S burner from Johnston Boiler Company.

All Johnston Boiler products are designed and manufactured to give excellent, efficient performance over a long period of time. All items supplied were chosen because of their ability to give high quality performance. Proper operating and maintenance procedures must be followed at all times to ensure a high degree of protection and safety. Operation of this equipment should not be considered free from all dangers inherent to operating a steam/hot water vessel with the firing of a combustible fuel.

The operator is solely responsible for properly maintaining the equipment. No amount of documented instruction can take the place of intelligent thinking and consistent safe practices. This manual is not intended to relieve the operator of the responsibility for proper operation.

The operator must be familiar with all components and operations of the burner. Identify and locate each component while reading this manual.

B. General Burner Description

The Johnston Series S burner is available in gas, oil, and combination gas and oil designs. Special designs are available for burning waste oil, tallow, and low-btu gas. Series S burners are engineered for firetube and watertube boilers ranging from 1800 to 3000 Boiler Horsepower (62,000 to 100,000 lbs Steam per hour).

All Johnston burners are factory test fired when installed on Johnston boilers.

C. Components of the Burner

Johnston Series S burners Fig 6.1 consists of a burner /plenum assembly and floor mounted blower.

Blower Assembly	Produces and controls the combustion air flow required for the burner.
Damper	(Includes blades, shafts, bearing and linkage arms, joints, and rods). Throttles the amount of combustion air being delivered to the burner.

Inlet Silencer (optional)	Insulated extension to reduce air entrance noise.
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Plenum Box Assembly	Collects the combustion air and fuel and directs it through the burner.
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Access Plate	Provides for access into burner internals.
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Refractory Sleeve	Supports combustion and helps establish aerodynamic flow field. The refractory sleeve also protects the boiler tube sheet from high temperature gases and radiation from the flame.
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Pilot Assembly	Provides a flame or high energy spark which is utilized to ignite the fuel at the main burner. Standard electrode setting is 1/8" to 3/16" gap.
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Scanner Assembly	Sensor which ensures that pilot and main flame have been established and are proven during all firing conditions. Either ultraviolet or infrared, depending on application.
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Oil Gun Assembly	Delivers and mixes the required amount of oil and atomizing air (primary air) to the burner, at the point of combustion, for oil firing. Small A series burners have one oil nozzle, while large A series burners have three individual oil nozzles.
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Main Gas Ring	Integral with refractory sleeve. Delivers the required amount of gas to be burned and distributes the gas at the point of combustion.
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Spuds or Plugs	Used to establish good mixing of the gas and combustion air at the burner front.
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Swirler	Controls the mixing of combustion air with the fuel at the burner front.
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Section 6:

Assembly Drawing of an S Burner

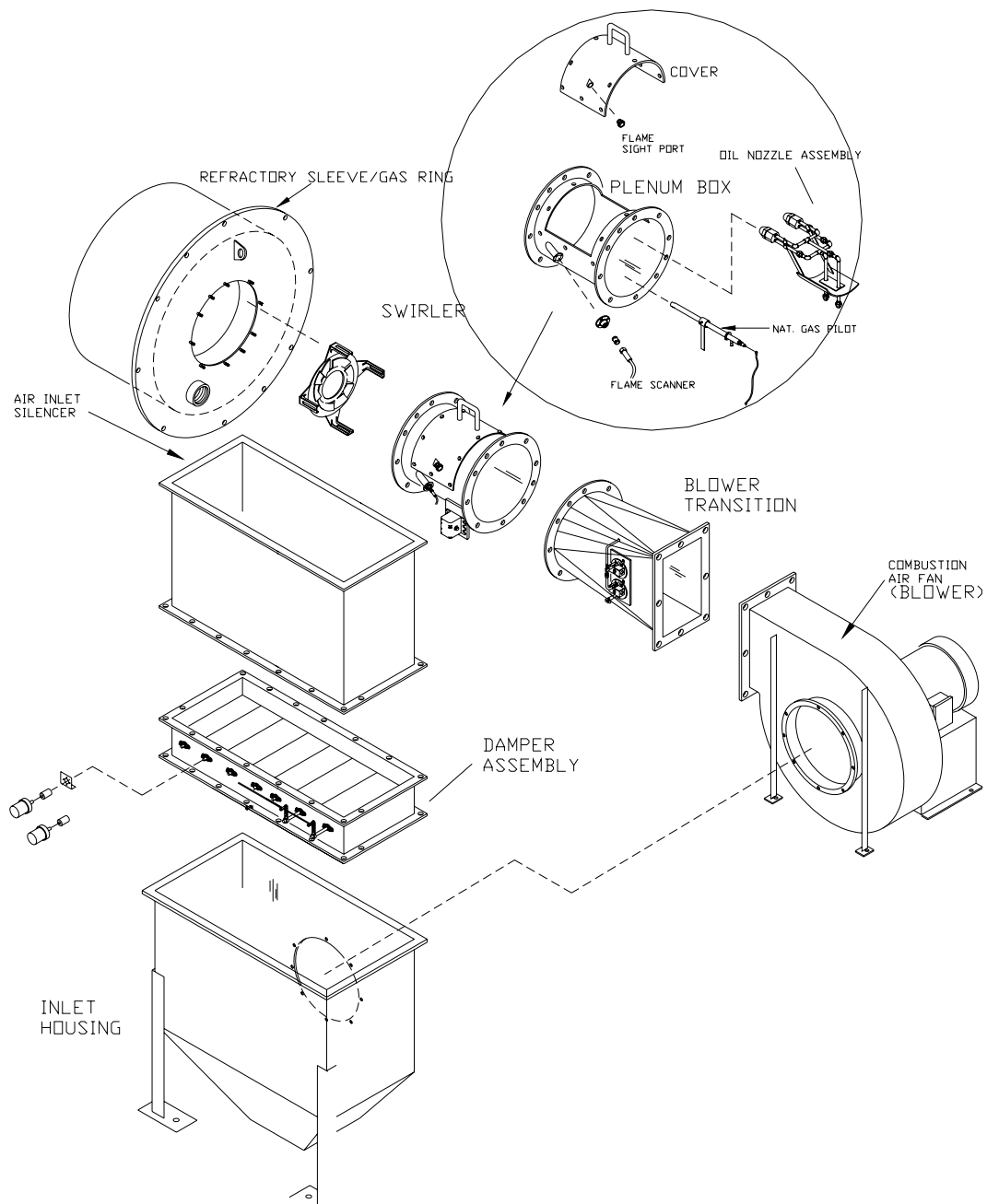


FIG 6.1

Section 6:

Burner Trim Components Fig 6.2

Firing Rate Motor	Electric Motor to rotate jackshaft on signal to increase or decrease burner firing rate.
Ignition Transformer	High voltage transformer , when energized, ignites pilot fuel.
Flame Scanner	Electronically recognizes the presence of flame.
Pilot	Normally natural gas or propane. Interrupted small flame to ignite main flame.

Jackshaft and Linkage Set Transfers movement from firing rate motor to fuel valves and air dampers.

Combustion Air Switch Permissive- Electric pressure switch to prove presence of combustion air.

Purge Air Switch Permissive- Electric pressure switch to prove the presents of air during burner purge cycle.

Section 6:

Assembly Drawing Burner Trim Components

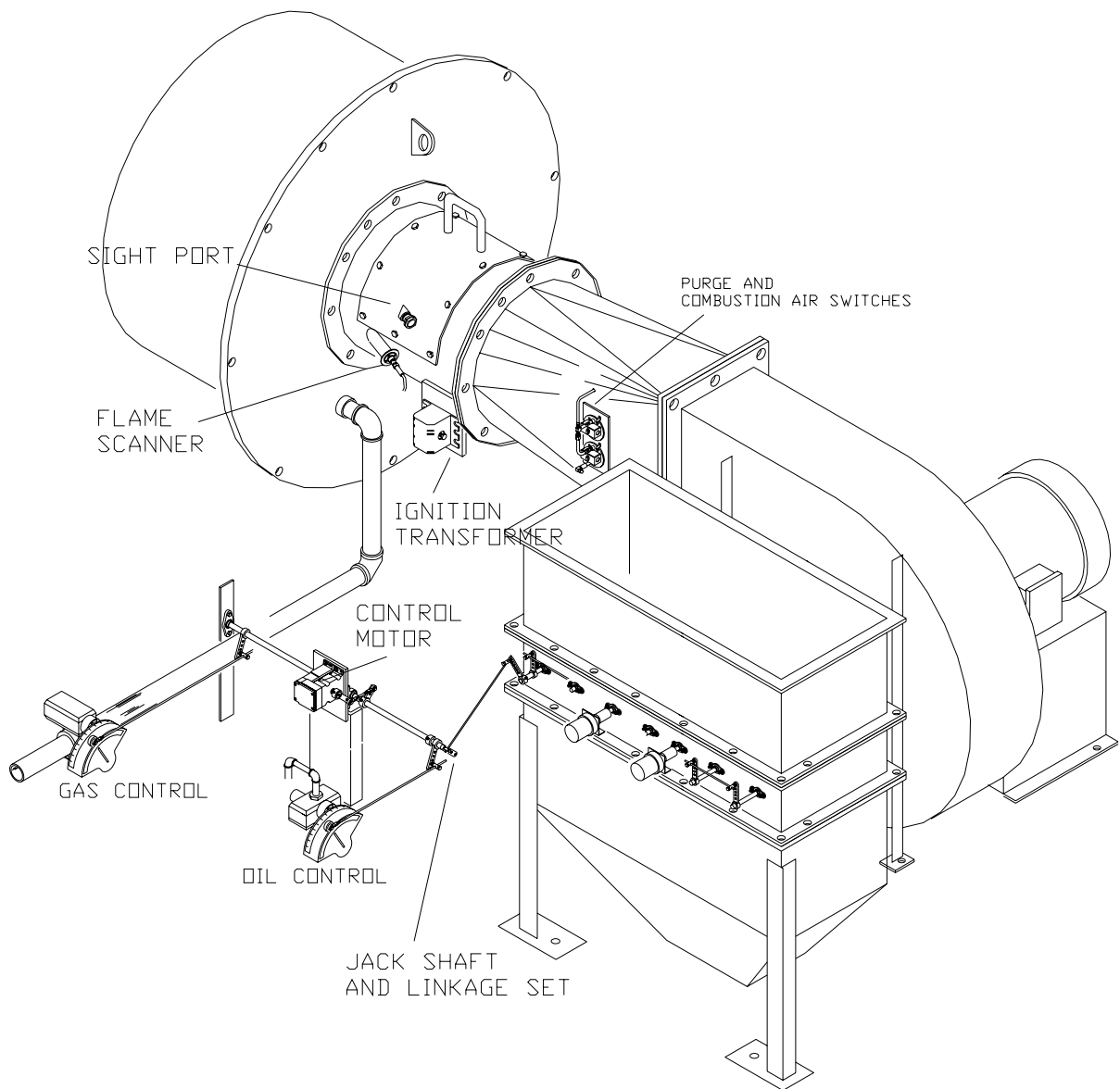


Fig. 6.2

Section 7: Fuel

A. Main & Pilot Gas Train

Pilot Gas Train Fig 7.1

Manual Shutoff Valve	Manual valve that controls the gas supply to the pilot train.
Pressure Regulator	Reduces the incoming gas supply pressure to meet the lower pressure requirements of the pilot.
Shutoff Valves	Electric normally closed valve(s), when energized, open to admit gas to the pilot. (One or two, depending on insurance/code requirements)
Pressure Gauge	Indicates the regulated gas supply pressure to the pilot assembly.
Vent Valve	Electric normally open valve, closes when energized. (when req'd) Relieves pressure between normally closed valves.

Low Gas Pressure Switch
Opens under a low gas supply pressure condition. Will not allow the burner to operate under this condition.

Safety Shutoff Valves
Open to admit gas to the main burner, but only after a pilot has been proven. Electrically actuated and equipped with a proof of closure switch. (If required)

Vent Valve
Electric normally open valve, closes when energized. (when req'd) Relieves pressure between normally closed valves.

Supervisor Shutoff Cock
Downstream shutoff cock, between main gas valve and flow control valve. Manually opens and closes the gas supply to the main burner.

Control Valve
Controls the amount of regulated gas supply pressure to the main burner. Valve is controlled by the burner management system.

Burner Pressure Gauge
Indicates the gas supply pressure to the main burner.

Main Gas Train Fig 7.1

Main Shutoff Cock	Upstream shutoff cock at entrance to the main gas train. Manual valve that controls the gas supply to the train.
Supply Pressure Regulator	Reduces the incoming supply pressure to meet the lower pressure requirements of the main burner and eliminates any fluctuation in the gas supply pressure.
Regulated Pressure Gauge	Indicates the regulated gas pressure to the main gas train.

High Gas Pressure Switch
Permissive- Contacts open under a high gas supply pressure condition. Will not allow the burner to operate in high gas pressure conditions.

Note: Gas supply for pilot should be upstream of main shutoff cock. (Certain main gas supply regulators require this connection to be downstream. Check piping schematics for correct connection).

Section 7:

Main and Pilot Gas Trains

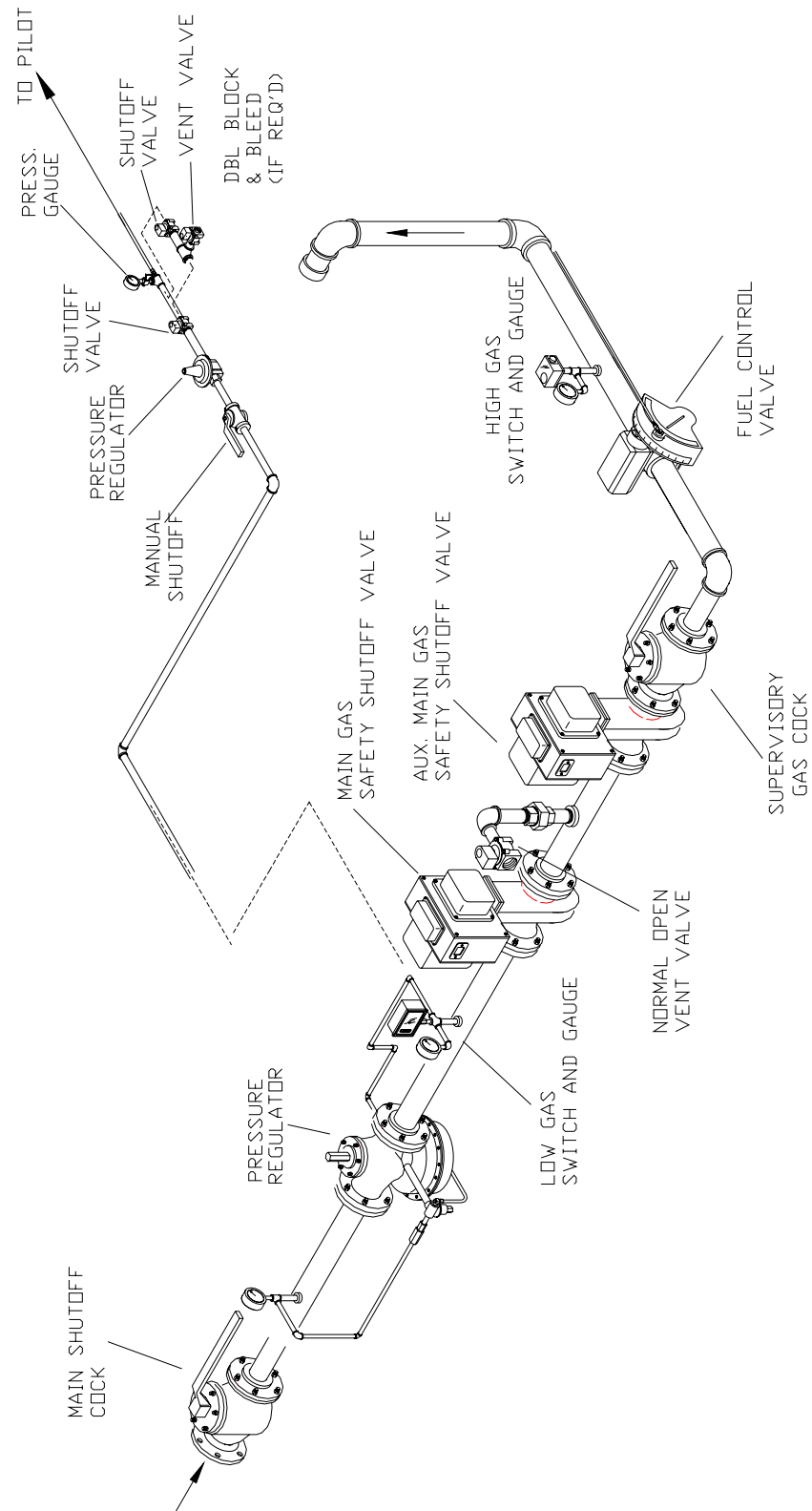


Fig. 7.1